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Bereken de volgende getallen in $\mathbb{C}$:

$$\frac{1}{i} - \frac{2}{3+i} + \frac{8}{3-i}$$

Oplossing:

$$\begin{aligned} & \frac{1}{i} - \frac{2}{3+i} + \frac{8}{3-i} \\ &= -i - \frac{2(3-i)}{(3+i)(3-i)} + \frac{8(3+i)}{(3-i)(3+i)} \\ &= -i - \frac{6-2i}{10} + \frac{24+8i}{10} \\ &= -i + \frac{-6+24}{10} + \frac{2i+8i}{10} \\ &= -i + \frac{18}{10} + \frac{10i}{10} \\ &= -i + \frac{9}{5} + i \\ &= \frac{9}{5} \end{aligned}$$

References